

Hydronic Pumps

Issued for Construction - Addendum 04

PROJECT	Mission Bay Data Center
EQUIPMENT	P-401 / P-402 chilled-water pumps
DESIGN AUTHORITY	Northline MEP Engineers
SPECIFICATION DATE	2026-05-18

1.01 Summary

Provide factory-assembled, base-mounted, end-suction centrifugal pumps with motors, bases, couplings, guards, seals, and accessories required for a complete chilled-water installation.

1.02 Submittals

Submit certified performance curves, dimensional drawings, motor data, connection sizes and locations, weights, installation instructions, minimum service clearances, and a point-by-point compliance statement. Product substitutions require written approval before purchase.

1.03 Quality Assurance

Pump and motor assembly shall be manufactured under an ISO 9001 quality system. Motors shall comply with NEMA MG 1 and meet NEMA Premium efficiency. Electrical components shall be suitable for 480 V, 3 phase, 60 Hz service.

2.01 Performance Requirements

Requirement	P-401 value	Tolerance / status	Evidence class
Design flow	420 gpm	No less than scheduled	Hard
Design head	105 ft	+/- 3 percent at duty point	Hard
Motor	30 hp, 480 V, 3 ph, 60 Hz	No overload at any curve point	Hard
Suction connection	4 in, ASME B16.1 Class 125 flange	Exact interface required	Hard
Discharge connection	3 in, ASME B16.1 Class 125 flange	Exact interface required	Hard
Working pressure	175 psig minimum	No reduction permitted	Hard
Efficiency	Minimum 79 percent at duty point	No lower than scheduled	Hard
Seal	Mechanical, carbon/ceramic, EPDM	Equivalent accepted	Negotiable
Finish	Manufacturer standard	Color may vary	Negotiable

2.02 Construction

Component	Requirement
Casing	Cast iron, back pull-out configuration, replaceable wear ring.
Impeller	Bronze, dynamically balanced, trimmed for scheduled duty point.
Shaft	Stainless steel or steel with replaceable sleeve.
Base	Common structural steel base sized for pump and motor assembly.
Coupling guard	OSHA-compliant, removable without disturbing alignment.
Bearings	Grease-lubricated, minimum L10 life of 100,000 hours at duty point.

2.03 Dimensional and Coordination Requirements

Constraint	Maximum / required value	Source
Housekeeping pad	2050 x 950 mm	M-601, Pump Room 4
Overall equipment envelope	1950 x 850 x 1150 mm	M-601 / coordination model
Motor-removal clearance	915 mm minimum behind motor	Manufacturer instruction
Electrical working clearance	914 mm clear in front of disconnect	E-401
Service aisle	760 mm minimum	M-601
Connection relocation	Maximum 50 mm with approved spool	Project coordination rule

3.01 Substitution Review

A proposed substitute will be evaluated on technical compliance, verified availability, total installed cost, schedule impact, physical fit, connection alignment, maintenance access, and installation access. Purchase price alone is not a basis for selection.

3.02 Required Approval Package

Provide a compliance matrix, certified curve, manufacturer dimensional drawing, current supplier quotation, inventory timestamp, delivery commitment, coordinated BIM object, clash report, maintenance-clearance review, cost impact, schedule impact, and draft RFI. Critical dimensions without manufacturer evidence shall be marked unresolved and shall not be inferred.